

Moving Charges and Magnetism

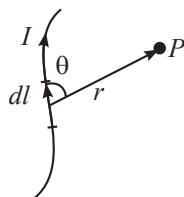
A static charge produces only electric field. A moving charge produces both electric field and magnetic field. A current carrying conductor produces only magnetic field.

Magnetic Field Produced by Current (Biot-Savart's Law)

The magnetic induction dB produced by an element dl carrying a current I at a distance r is given by:

$$dB = \frac{\mu_0 \mu_r}{4\pi} \frac{I dl \sin \theta}{r^2} \Rightarrow \vec{dB} = \frac{\mu_0 \mu_r}{4\pi} \frac{I(d\vec{l} \times \vec{r})}{r^3}$$

Here, the quantity $I dl$ is called as current element.



μ = permeability of the medium = $\mu_0 \mu_r$

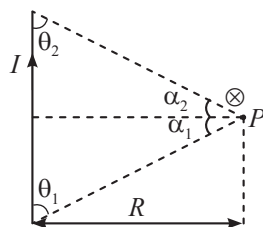
μ_0 = permeability of free space

μ_r = relative permeability of the medium (Dimensionless quantity)

Unit of μ_0 and μ is N A^{-2} or H m^{-1} ; $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$

Magnetic Induction Due to a Straight Current Conductor

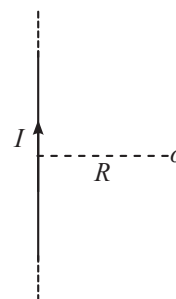
(i) Magnetic induction due to a finite wire.



$$B = \frac{\mu_0 I}{4\pi R} (\cos \theta_1 + \cos \theta_2) = \frac{\mu_0 I}{4\pi R} (\sin \alpha_1 + \sin \alpha_2)$$

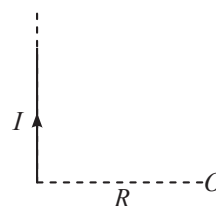
(ii) Magnetic induction due to an infinitely long wire

$$B = \frac{\mu_0 I}{2\pi R} \otimes (\alpha_1 = 90^\circ; \alpha_2 = 90^\circ)$$



(iii) Magnetic induction due to semi infinite straight conductor

$$B = \frac{\mu_0 I}{4\pi R} \otimes (\alpha_1 = 0^\circ; \alpha_2 = 90^\circ)$$



Magnetic Field Due to a Flat Circular Coil Carrying a Current

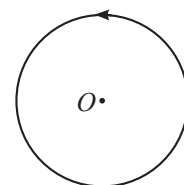
(i) At its centre: $B = \frac{\mu_0 NI}{2R} \odot$

where

N = total number of turns in the coil

I = current in the coil

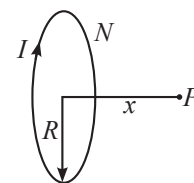
R = Radius of the coil



(ii) On the axis: $B = \frac{\mu_0 NI R^2}{2(x^2 + R^2)^{3/2}}$

where x = distance of the point from the centre.

It is maximum at the centre, $B_c = \frac{\mu_0 NI}{2R}$



(iii) Magnetic field due to flat circular arc at its centre:

$$B = \frac{\mu_0 I \theta}{4\pi R}$$



Magnetic Field Due to Infinite Long Solid Cylindrical Conductor of Radius R

❖ For $r \geq R$: $B = \frac{\mu_0 I}{2\pi r}$

❖ For $r < R$: $B = \frac{\mu_0 I r}{2\pi R^2}$

Magnetic Induction Due to a Solenoid

$B = \mu_0 n I$, where n is number of turns per meter and I is current. Direction is along the axis.

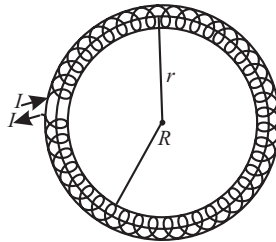
Magnetic Induction Due to Toroid

$B = \mu_0 n I$

Where $n = \frac{N}{2\pi R}$

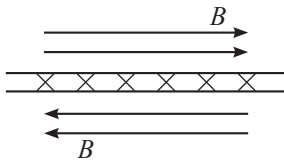
R is radius of toroid

N is total turns and $R \approx r$



Magnetic Induction Due to Current Carrying Sheet

$B = \frac{1}{2} \mu_0 \lambda (\lambda = \text{Linear current density (A/m)})$



Ampere's Circuital Law

$\oint \vec{B} \cdot d\vec{l} = \mu_0 \Sigma I$ where ΣI = algebraic sum of all the enclosed current.

Motion of a Charge In Uniform Magnetic Field

(a) When $\vec{v} \parallel \vec{B}$: Motion will be in a straight line and $\vec{F} = 0$

(b) When $\vec{v} \perp \vec{B}$: Motion will be in circular path with radius

$R = \frac{mv}{qB}$ and angular velocity $\omega = \frac{qB}{m}$ and $F = qvB$.

(c) When $\vec{v}$ is at angle θ ($\theta \neq 0, \pi, \pi/2$) to $\vec{B}$: Motion will be helical with radius

$R = \frac{mv \sin \theta}{qB}$ and pitch $P_H = \frac{2\pi mv \cos \theta}{qB}$ and $F = qvB \sin \theta$.

Lorentz Force

An electric charge ' q ' moving with a velocity $\vec{v}$ through a magnetic field of magnetic induction $\vec{B}$ experiences a force $\vec{F}$, given by $\vec{F} = q \vec{v} \times \vec{B}$. There fore, if the charge moves in a space where both electric and magnetic fields are superposed.

$\vec{F} = \text{net electromagnetic force on the charge} = q\vec{E} + q\vec{v} \times \vec{B}$

This force is called the Lorentz Force

Motion of Charge in Combined Electric Field and Magnetic Field

❖ When $\vec{v} \parallel \vec{B}$ and $\vec{v} \parallel \vec{E}$, motion will be uniformly accelerated in straight line as $F_{\text{magnetic}} = 0$ and $F_{\text{electrostatic}} = qE$

So the particle will be either speeding up or speeding down

❖ When $\vec{v} \parallel \vec{B}$ and $\vec{v} \perp \vec{E}$, motion will be uniformly accelerated in a parabolic path

❖ When $\vec{v} \perp \vec{B}$ and $\vec{v} \perp \vec{E}$, the particle will move undeflected and undeviated with same uniform speed if $v = \frac{E}{B}$ (This is called as velocity selector condition)

Magnetic Force on a Straight Current Carrying Wire

$\vec{F} = I (\vec{L} \times \vec{B})$

I = current in the straight conductor

L = displacement between end points of the conductor in the direction of the current in it

$\vec{B}$ = magnetic induction (Uniform throughout the length of conductor)

Note: In general, force is $\vec{F} = \int I(d\vec{l} \times \vec{B})$

Magnetic Interaction Force Between Two Parallel Long Straight Current wires

The interaction force between 2 parallel long straight wires is:

(i) Repulsive if the currents are anti-parallel.

(ii) Attractive if the currents are parallel.

This force per unit length on either conductor is given by

$F = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$

Where r = perpendicular distance between the parallel conductors

Magnetic Torque on a Closed Current Loop Kept in a Uniform Magnetic Field

When a plane closed current circuit of ' N ' turns and of area ' A ' per turn carrying a current I is placed in uniform magnetic field, it experience a zero net force, but experiences a torque given by

$\vec{\tau} = NI \vec{A} \times \vec{B} = \vec{M} \times \vec{B} = BINA \sin \theta$

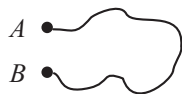
where $\vec{A}$ = area vector outward from the face of the loop where

the current is anticlockwise, B = magnetic induction of the uniform magnetic field and

$\vec{M}$ = magnetic moment of the current loop = $NI \vec{A}$



Force on a Random Shaped Conductor in a Uniform Magnetic Field



- ❖ Magnetic force on a closed loop in a uniform $\vec{B}$ is zero
- ❖ Force experienced by a wire of any shape is equivalent to force on a wire joining points A and B in a uniform magnetic field.

Magnetic Moment of A Revolving Charge

If a charge q is revolving at an angular velocity ω , its equivalent current is given as $I = \frac{q\omega}{2\pi}$ and its magnetic moment is $M = I\pi R^2 = \frac{1}{2}q\omega R^2$.

